

Social Psychology and Its Methods

Study Goal

1) What is social psychology?
2) What do social psychologists study?
3) Basic concepts of social psychology
4) What methods do we use to test theories?
5) Replication crisis

Questions?

Why are people inclined to stereotype members of different groups?
Why do people risk their lives to help others?
Why do some marriages flourish and others fail?
How do orderly crowds turn into violent mobs?

Social Psychology

- Psychology:
  - The scientific study of behavior and the mind

- Social Psychology - Gordon Allport, 1985 :
  - The scientific study of the feelings, thoughts, and behaviors of individuals in social situations.



Characterizing Social Psychology

Social psychology

- The scientific study of the feelings, thoughts, and behaviors of individuals in social situations

The importance of the ABCs to social psychology

- Affect (feelings)
- Behavior (actions)
- Cognitions (thoughts)

How does social psychology differ from:

- Sociology?
  - It's about the influence of social factors on behavior
- Other areas of psychology?
  - Cognitive psychology
    - It's social
  - Clinical psychology
    - It's about normal populations
  - Personality psychology
    - It's about people in general
    - (psychological processes that people have in common with one another)

Why is social psychology important?



Research (and Theory) in Social Psychology

- Theory
  - A set of related propositions intended to describe some phenomenon or as pect of the world.

Signs of a Good Theory

- Posits causal relationships (explains observations)
- Attempts to be coherent (organizes observations)
- Tells a good story
- Aims for parsimony (explains a range of observations with a few basic principles)
- Is testable (provides direction for research)
- Proves fertile (generates new questions)
- Solves problems (has practical value)

Susan Fiske (2004)

Research (and Theory) in Social Psychology

- Theory
  - A set of related propositions intended to describe some phenomenon or aspect of the world.
- Create hypotheses based on a theory
  - Testable predictions about the relationship between two or more variables
  - If-then statements about how variables are related; what will happen under particular circumstances
- Test the hypotheses
- Theory refinement (mechanisms, boundary conditions, extensions)

Example: Ego depletion

- *Does exerting willpower or control deplete us?*

Theory:

- Ego depletion: The active self is a limited resource
  - People have a common source of controlled resources/willpower from which they draw.
  - Each time we try to control our will, we reduce this pool of resources.

Baumeister et al. (1998)

How Can We Test This Theory?

- Create a hypothesis:
  - Women on a diet have fewer cognitive resources than women not on a diet.
- Method
  - Correlational
    - Measure natural associations to assess the relationship between two or more variables

Correlational Research

- Examine whether the occurrence of A is related to the occurrence of B
  - Variable A – diet or not
    - How to measure this?
    - One way - Restrained Eating Scale
  - Variable B – cognitive resources
    - How to measure this?
    - One way – Stroop Task

Correlational Research

- Examine whether the occurrence of A is related to the occurrence of B.
- Measure 2 or more variables and assess whether there is a relationship between them.

Measure of Diet - Restrained Eating Scale

1. How often do you diet?

Never Rarely Sometimes Usually Always

2. What is the maximum amount of weight (in pounds) that you have ever lost in one month?

0 4 5-9 10-14 15- 19 20+

3. Do you have feelings of guilt after overeating?

Never Rarely Often Always

4. How conscious are you of what you are eating?

Not at all Slightly Moderately Very Much

Measure of Cognitive Resources - Stroop Task

黄色

Stroop Task

黄色

Stroop Task

蓝色

Stroop Task

红色

Correlational Research

- Hypothesis:
  - Women on a diet have fewer cognitive resources than women not on a diet.
- Results:
  - Higher scores on dieting scale related to higher Stroop interference (slower color naming).

- What is another way of testing this relationship?
- Another way of operationalizing willpower and controlled resources?

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How Can We Test This Theory?

- Create a hypothesis:
  - People who suppress their emotions have less willpower over actions compared to people who do not suppress their emotions.
- Method
  - Experimental
    - Manipulate one variable to see if it affects another variable in a predicted fashion

Experimental Research

- Examine whether Variable A causes changes in Variable B
  - Manipulate independent variable (assumed causes of phenomena)
  - Observe effect on dependent variable
- Basic principles:
  - Random Assignment
    - Participants have an equal chance of being in any of the conditions

Experimental Research

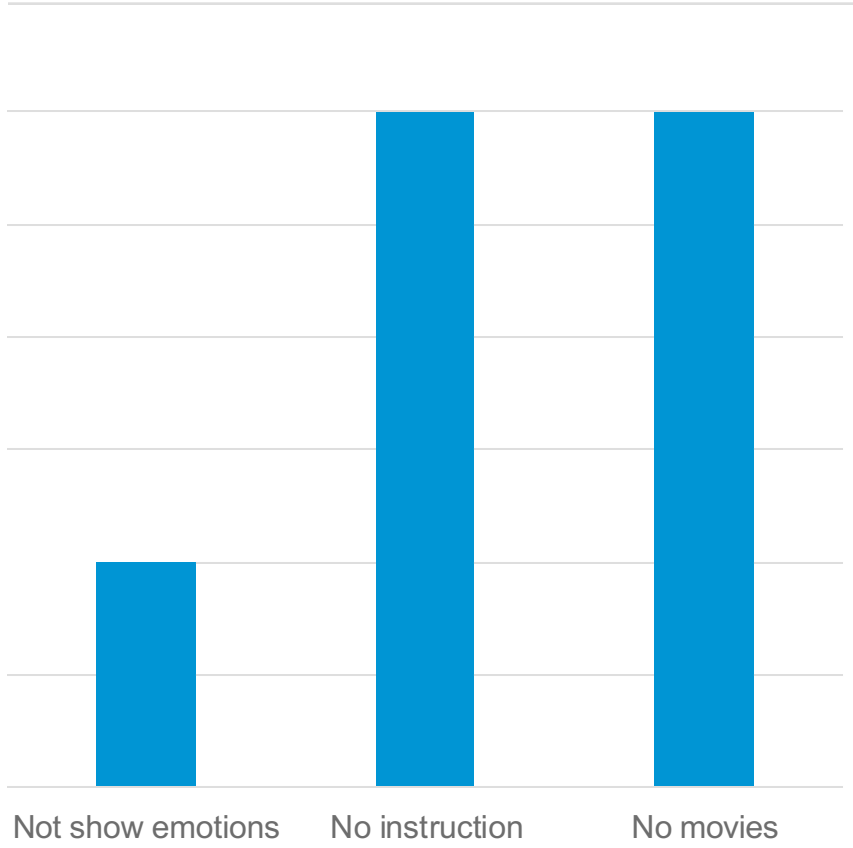
- Independent Variable (manipulated variable)
  - 1/3 subjects told not to show emotions during sad movie
  - 1/3 subjects given no instructions during sad movie
  - 1/3 subjects not shown a movie and no instructions

- Dependent Variable (observed effect)
  - All subjects hold hand in cold water for as long as possible

Experimental Research

- Hypothesis: People who suppress their emotions have less willpower over actions compared to people who do not suppress their emotions.

- Results: People who were told not to show emotions during sad movies hold their hands shorter compared to people who were given no instructions or people who were not given a movie.



Types of Validity

Validity issues are related to whether a relationship exists between 2 variables.

- Internal Validity
  - whether changes in the dependent variable were caused by changes in the independent variable
- Construct Validity
  - whether the manipulation of the independent variable and the measurement of the dependent variable are good representations of the theoretical constructs
- External Validity
  - whether the results generalize to other labs, participants, settings (e.g., in the field)

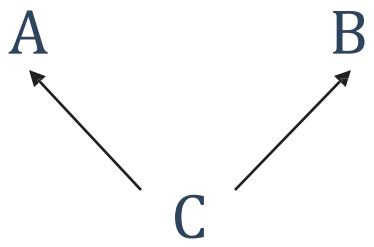
Possible Problems with Correlational Research

Issues with internal validity:

- Reverse causality problem



- Third variable problem



Possible Problems with Experimental Research

Issues with external validity:

Don’t know whether results generalize.

Possible Problems with Correlational & Experimental Research

Issues with construct validity:

Whether operationalization and measurement of variables reflect the theoretical construct.

Other Methods

- Observational Research:
  - just looking at a phenomenon in a systematic way, with a view to understan ding what's going on and coming up with hypotheses about why things are as they are
- Archival Research:
  - researchers can look at evidence found in *archives* of various kinds, such as databases containing ethnographic (anthropological) descriptions of people in different cultures
- Surveys:
  - can be conducted using either interviews or written questionnaires

Other Methods

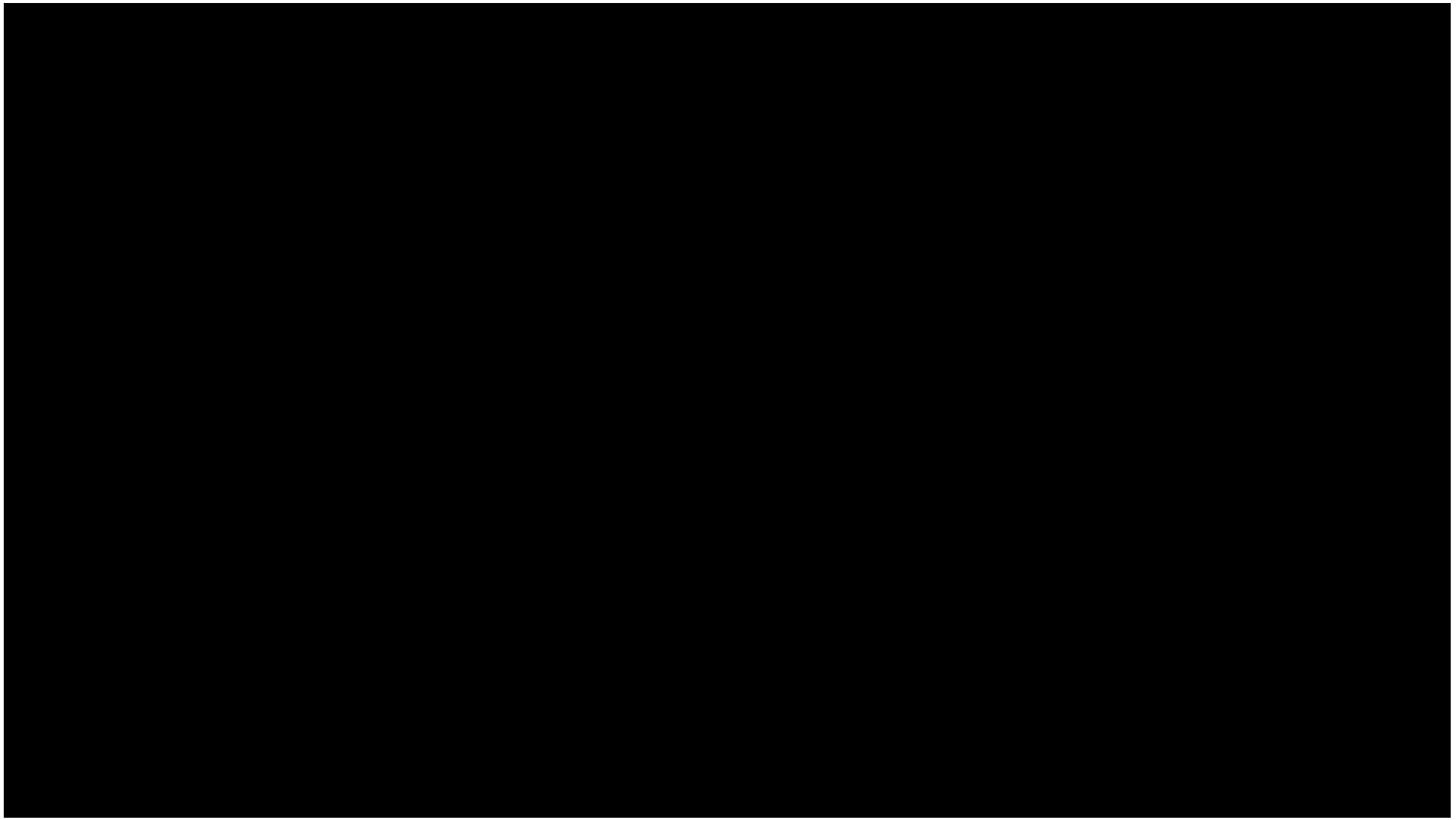
- Meta-Analysis Research:
  - A statistical technique that combines the results of multiple studies to determine overall trends or effects in a particular area of research.
- Big Data Research
  - Big data research involves analyzing large-scale datasets, often collected from digital platforms, social media, or online transactions, to uncover patterns, trends, and associations in human behavior. Social psychologists use big data to study large-scale social interactions and behaviors that would be difficult to observe through traditional methods.

Replication Crisis

- Who believes in ESP?
  - Extra sensory perception – a psychic ability to read other's minds or the future

https://www.youtube.com/watch?v=42QuXLucH3Q

Replication Crisis



Replication Crisis

- You may also find this video interesting: https://www.youtube.com/watch?v=vBzEGSm23y8&fbclid=IwAR0nlV9FzQDMF4vpu8gyzgKD4Rd36a3VpdHGGQ-xwrF7YCvp5Vca_abVRwE

Are Ego Depletion Effects Real?

- Does exerting willpower or control deplete us?
- Do the effects replicate? https://www.slate.com/articles/health_and_science/cover_story/2016/03/ego_depletion_an_influential_theory_in_psychology_may_have_just_been_debunked.html



COVER STORY | READ THIS FIRST. | MARCH 6 2016 8:02 PM

Everything Is Crumbling

An influential psychological theory, borne out in hundreds of experiments, may have just been debunked. How can so many scientists have been so wrong?

By Daniel Engber

Causes of Nonreplication

- Small sample sizes
- p-hacking/questionable research practices (QRPs) to ensure ($p < .05$)
  - dropping subjects
  - dropping conditions
  - dropping dependent variables
  - file drawers – filled with studies that did not work

Improving Science

- Increase sample sizes
- Direct (vs. Indirect) replications by multiple labs
- Preregistration of studies to prevent p-hacking
- Offsite repositories for stimuli (to facilitate replication attempts)
- Offsite repositories to store data

The Evolution of a Theory

- Science is a cumulative process: continually evolving moving toward a more useful and hopefully valid understanding of reality.
- It depends on direct and conceptual replications.

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